Experiment No: 01

Name of Experiment: Familiarization of different Equipment involved with Electronics Sessional.

Objectives:

1. To learn about the commonly used equipment in the Electronics Sessional.
2. To observe the various parameter of the electric circuit.
3. To observe the different instrument which are usually need for measurement.
4. To understand the use of the instrument and components in experiment.

Theory:

Electronics is a scientific and engineering discipline that studies and applies the principles of physics to design, create, and operate devices that manipulate electrons and other electrically charged particles. It is a subfield of physics and electrical engineering which uses active devices such as transistors, diodes, and integrated circuits to control and amplify the flow of electric current and to convert it from one form to another, such as from alternating current (AC) to direct current (DC) or from analog signals to digital signals. Electronics is often contrasted with electrical power engineering, which focuses on generation, transmission, and distribution of electric power rather than signal processing or device level control.

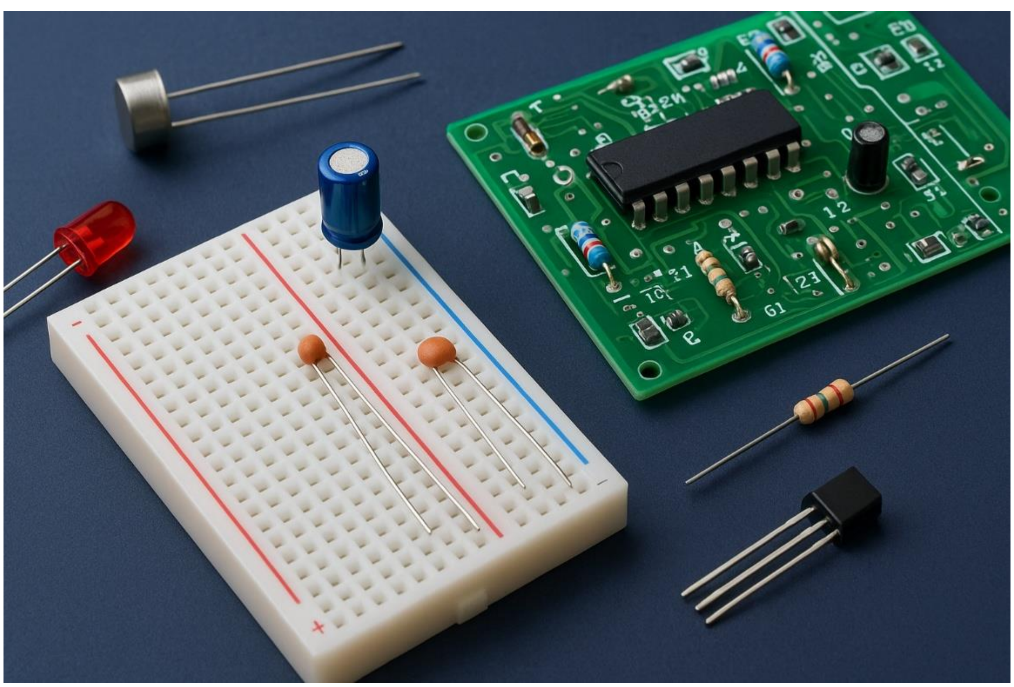

Figure 1.01: Electronics

Major component used equipment in the Electronics Sessional:

Resistor:

A resistor is a two terminal electric component. Generally, it is used to reduce the current flow. It is also used to many others purposes such as to adjust signal levels, to divide voltage, to terminate transmission line. High-power registers can dissipate many watts of current. The resistance is usually measured in ohms. The symbol is called omega. Resistors values are normally shown in color, hand bands. Each color represents different value.

	Color	Signficant figures			Multiply	Tolerance (%)	Temp. Coeff. (ppm/K)	Fail Rate (%)
Bad	black	0	0	0	x 1		250 (U)	
Beer	brown	1	1	1	x 10	1 (F)	100 (S)	1
Rots	red	2	2	2	x 100	2 (G)	50 (R)	0.1
Our	orange	3	3	3	x 1K		15 (P)	0.01
Young	yellow	4	4	4	x 10K		25 (Q)	0.001
Guts	green	5	5	5	x 100K	0.5 (D)	20 (Z)	
But	blue	6	6	6	x 1M	0.25 (C)	10 (Z)	
Vodka	violet	7	7	7	x 10M	0.1 (B)	5 (M)	
Goes	grey	8	8	8	x 100M	0.05 (A)	1(K)	
Well	white	9	9	9	x 1G			
Get	gold			3th digit only for 5 and 6 bands	x 0.1	5 (J)		
Some	silver				x 0.01	10 (K)		
Now!	none					20 (M)		

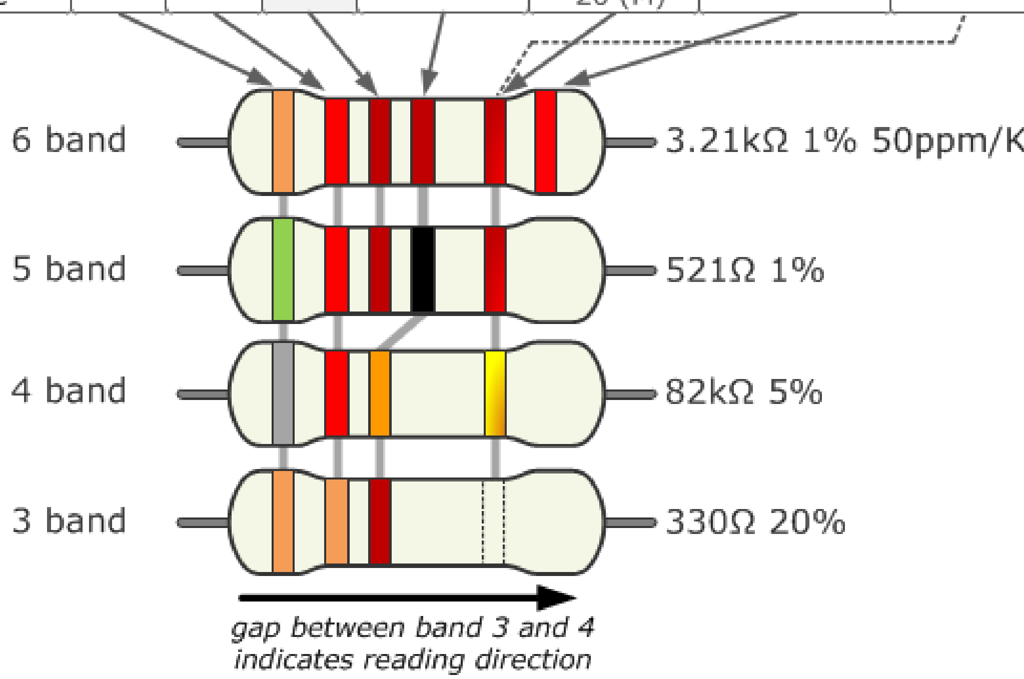

Figure 1.02: Resistor

Multimeter:

A multimeter is an important instrument for electric measurement. It can measure current flow and voltage both. It can also measure resistance. A multimeter also known as volt-ohm meter. It can measure various electrical properties because its unit is equipped with voltmeter ameter and ohm-meter functionality. In case of using multimeter as a volt meter, firstly we have to set the dialer in the voltage range. Then we have to connect the voltmeter with the

resistance in parallel pathway and the value will be shown in display. In case of using multimeter as an ameter, firstly, we have to set the dialer in the desired range of ampere. Then we have to connect the connectors of voltmeter with the resistance in parallel pathway with the resistor and the value of the current flow will be shown in display.

Figure 1.03: Multimeter

Project board/Breadboard:

A breadboard is solderless basic circuit board. It is used to make temporary circuit of prototype of electric circuit. It doesn't require any soldering in case of circuit making. We can easily attach any component in breadboard. For this reason, breadboard is widely popular.

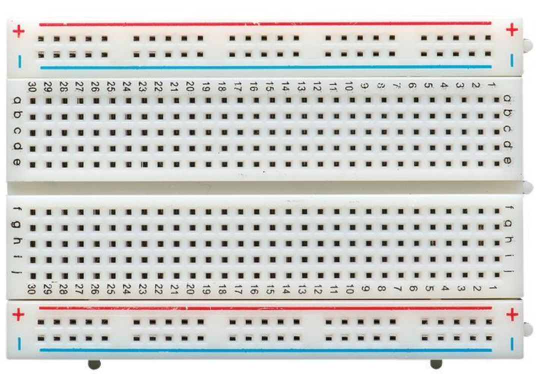

Figure 1.04: Project board/Breadboard

Capacitor:

A capacitor is an electronic component that stores and releases electrical energy in an electric field. It consists of two conductive plates separated by an insulator called a dielectric.

Figure 1.05: Capacitor

Diode:

A diode is a two-terminal electronic component that allows electric current to flow primarily in one direction, acting like a one-way switch for electricity.

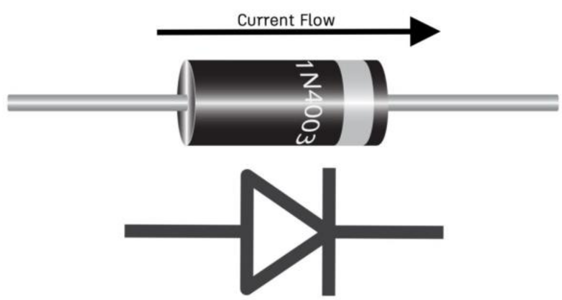

Figure 1.06: Diode

Zener diode:

A Zener diode is a special type of diode designed to operate in reverse bias, allowing current to flow in the opposite direction when the voltage across it reaches a specific "Zener voltage".

Figure 1.07: Zener diode

LED:

A light-emitting diode (LED) is a semiconductor device that emits light when current flows through it. Electrons in the semiconductor recombine with electron holes, releasing energy in the form of photons. The color of the light (corresponding to the energy of the photons) is determined by the energy required for electrons to cross the band gap of the semiconductor. White light is obtained by using multiple semiconductors or a layer of light-emitting phosphor on the semiconductor device.

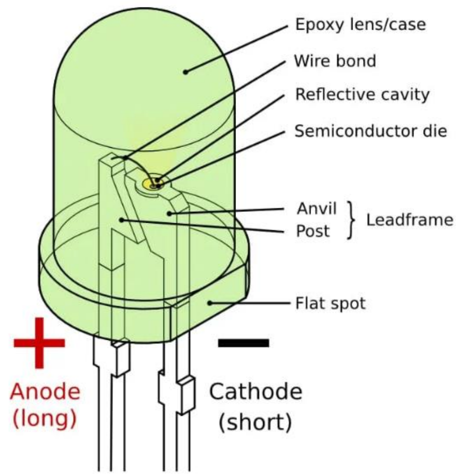

Figure 1.08: LED

Veroboard:

Veroboard is a type of stripboard, electronics prototyping board made of a single-sided, copper-clad material with a grid of holes.

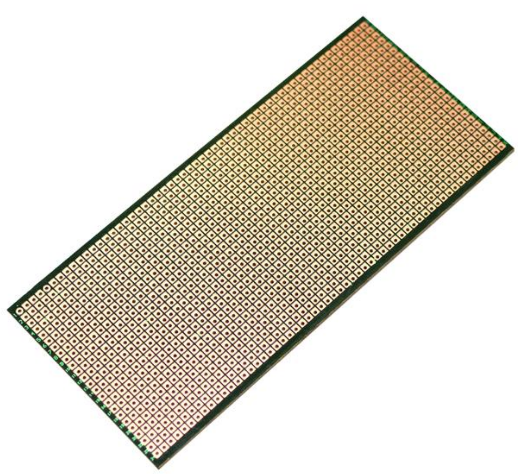

Figure 1.09: Veroboard

Oscilloscope:

An oscilloscope is an electronic test instrument that graphically displays varying voltages as a function of time, showing the shape of an electrical signal.

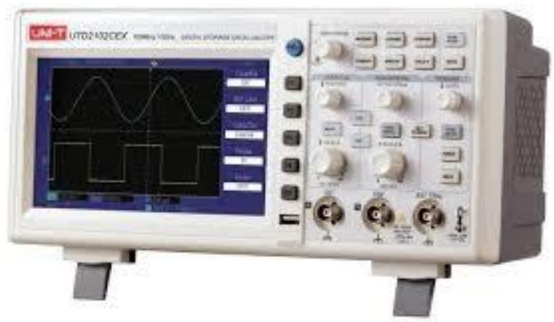

Figure 1.10: Oscilloscope

Op-Amp:

An op-amp, or operational amplifier, is a high-gain, DC-coupled voltage amplifier with a differential input and a single-ended output.

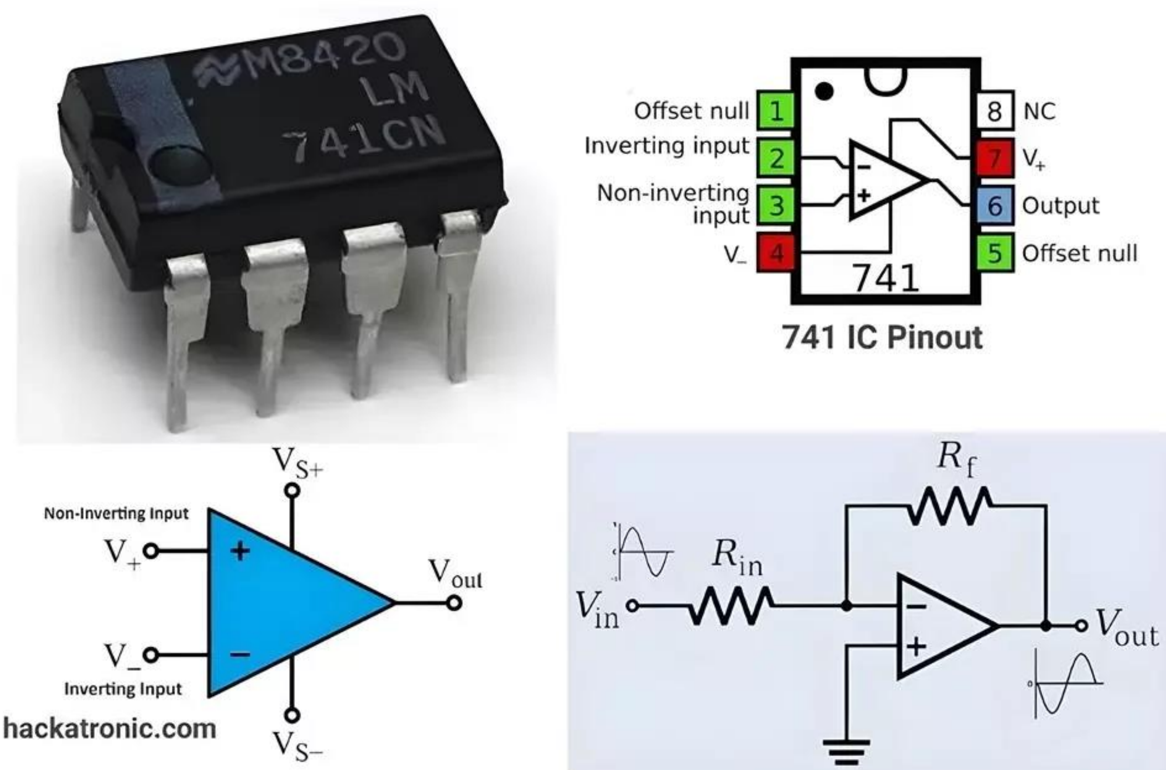

Figure 1.11: Op-Amp

BJT:

BJT stands for Bipolar Junction Transistor, a three-terminal semiconductor device that uses both electrons and holes as charge carriers.

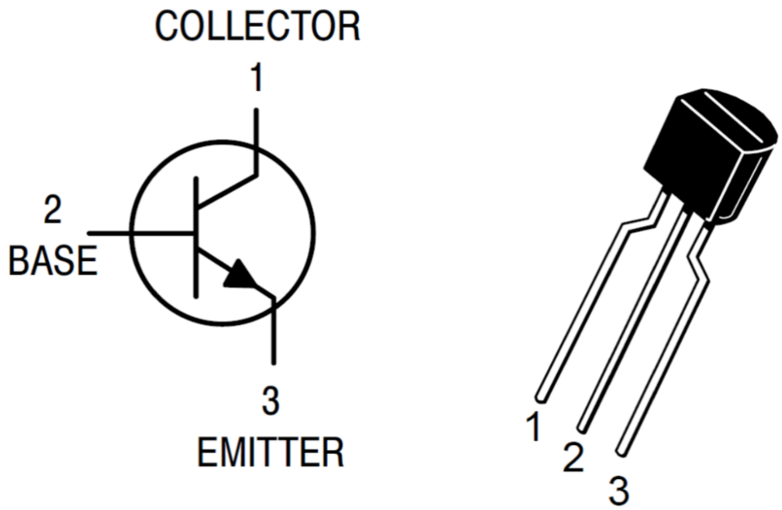

Figure 1.12: BJT

MOSFET:

A MOSFET, or metal-oxide-semiconductor field-effect transistor, is a semiconductor device that acts as an electrical switch or amplifier, controlling the flow of current between two terminals based on the voltage applied to a third terminal.

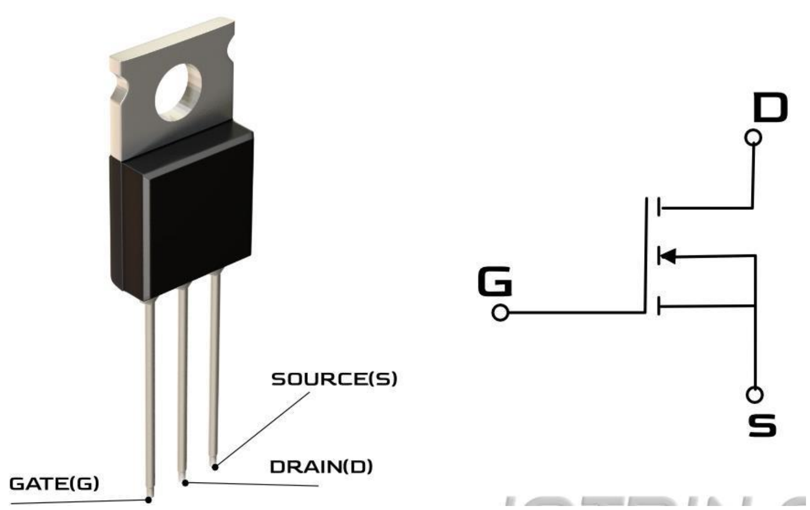

Figure 1.13: MOSFET

Soldering iron:

A soldering iron is a hand tool with a heated tip used to melt solder to join metal parts, wires, or electronic components. It consists of a heated metal tip, called the "bit," and an insulated handle, and is typically powered by electricity to generate high temperatures needed to melt the solder.

Figure 1.14: Soldering iron

Procedure:

1. Observe each electronic component demonstrated in the lab.
2. Carefully hold and examine each component by hand.
3. Note down the IC number (if applicable).
4. Understand the working principle and other important features of each component clearly.

Discussion:

Conclusion:

Reference (If any):